

ACCELERATION READOUTS IN HQIV FLYBY ANOMALIES, SPARC DISKS, AND LONG-TERM GALAXY–FILAMENT EVOLUTION

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ABSTRACT. **Position in the publication chain.** This is the twelfth paper in the HQIV preprint series (Tier-3 applied phenomenology). Tier 0–1 fix the O-Maxwell spine [?, ?, ?, ?, ?]; the longitudinal–coronal pair [?, ?] documents boundary-oriented phase-gradient stress on wires and flux tubes. The present note applies the same horizon-repartition picture to flybys, SPARC rotation curves, WHIM filament seeding, and long-term disk–filament coevolution (wide binaries remain a separate low-acceleration regime).

Background. Anderson et al. [?] reported millimetre-per-second discrepancies between Doppler tracking and classical Newton+ J_2 models for several Earth gravity assists. Among the four most quoted cases, two are clean (NEAR-Shoemaker 1998, Galileo I 1990), one is contaminated by thruster activity within an hour of closest approach (Cassini 1999), and one (Rosetta-I 2005) sits above the atmosphere with no agreed-upon classical explanation.

Approach. HQIV adds a direction-dependent inertia screen $f(a, \varphi) = a/(a + \varphi/6)$ with horizon repartition ($\varphi_{\text{eff,part}}$ on the geodesic, $\varphi_{\text{eff,full}}$ on the metric channel), a derived Lense–Thirring release fraction $\lambda = \gamma \sin^2 \theta \rho_{\text{pol}}$ ($\gamma = 2/5$), and a co-spinning mass-horizon Doppler lapse ϵ_{spin} in the metric channel only—the same structural split as the boundary φ jump in [?]. Reproducers: orbital flyby scaffold and `scripts/hqiv_orbital_flyby_omaxwell.py`; SPARC import via `scripts/hqiv_sparc_rotation.py` (WHIM on by default), `scripts/hqiv_whim_filament.py`, `scripts/hqiv_filament_crossmatch.py`, and `scripts/hqiv_spin_alignment.py`.

Solver. Classical RK4 with corrected (1, 2, 2, 1)/6 weights, verified to fourth-order on a Kepler benchmark, with Sun and Moon positions advanced *per substage* from the encounter epoch. The HQIV minus classical excess is stable to $\lesssim 0.01$ mm/s across $\Delta t \in [0.5, 4]$ s.

Result. At the nominal coupling (horizon repartition + φ_{hom} geometric readout, no extra chart κ scales) the calculator delivers, for the four catalog Earth flybys (literature Doppler residual in parentheses): NEAR +1.84 mm/s (+13.46), Galileo I +3.03 mm/s (+3.92), Cassini +0.24 mm/s (−0.50, thruster-contaminated; excluded from goodness), Rosetta-I −1.77 mm/s (+1.80 ± 0.05, sign open; §11.4). The derived λ is the dominant single change versus the legacy scalar lapse ($\lambda = 0.5$ fixed, NEAR +3.93 mm/s). Dropping L–T (isotropic horizon only) gives NEAR −0.87 mm/s, confirming the oriented metric channel is load-bearing. The spin input is not a fitted Anderson-style scalar: it is the Doppler projection of the rotating mass horizon onto the spacecraft velocity, suppressing radial and polar passages relative to equatorial tangential passages. No PDG mass or fitted ephemeris is used. Flyby, galaxy, and wide-binary readouts share the homogeneous $\varphi_{\text{hom}} \approx 2cH_0$ geometric lapse (Kirchhoff propagation-shell band; §13, §14), not the hadron lock-in shell **referenceM** = 4. On the full SPARC catalog [?] with gas+stellar+bulge components, a smooth WHIM filament seed/active blend replaces the homogeneous cosmic floor at small radius for diffuse systems (no halo fit). At fiducial Υ and no per-galaxy knobs, HQIV beats baryonic-only Newton on **106**/153 filtered galaxies ($\Sigma\chi^2$ ratio 0.31 vs 0.39 without WHIM; median $\chi^2_{\text{red}} = 34.8$ vs 49.9; $Q \leq 2$, $i \geq 30^\circ$). DDO154 drops from $\chi^2_{\text{red}} \approx 685$ to ≈ 118 . Carrón+2022 filament spines [?] cross-match 115/175 SPARC fields (VizieR sky positions, 2.25° match; 4.5° at $|\text{Dec}| \geq 55^\circ$). Coupled torque exchange on the WHIM boundary is reported as a σ_8 diagnostic (spin alignment over Gyr) and does *not* enter the instantaneous φ readout. Diffuse late-type galaxies (Im/Sm/BCD/DDO) remain the stress test; regenerate Figure 1 with `papers/orbital_flyby/scripts/generate_sparc_map.py`.

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. Fin 4), and

horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics. In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (e_1, e_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure` / `Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \text{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. SCOPE AND CLAIM DISCIPLINE

This is a falsifiable model and a reproducible set of calculators, not a proof that acceleration anomalies are HQIV in origin. We make the following claim discipline explicit:

- (1) **Locked equations.** A single LaTeX equation sheet (this manuscript, `scripts/hqiv_flyby_equations.py`, and the Lean scaffold) is the only source of truth. CLI: `--equations latex` prints the same sheet used here.
- (2) **Per-mission honesty.** We only score the integrator against missions whose literature residual is plausibly clean (NEAR, Galileo I). Cassini’s -0.5 mm/s is well below its thruster injection budget and is reported but not scored. Rosetta-I’s sign mismatch is flagged as an open item. Galileo’s 1992 flyby (GLI-2) at ~ 300 km altitude is atmospheric-drag dominated and is not included.
- (3) **No PDG injection.** The horizon shell ladder $\varphi(m) = 2(m+1)$ and `referenceM = 4` appear in the hadron/mass corpus only. Solar-system acceleration readouts (flyby, SPARC, wide binaries) use φ_{hom} and radial geometry, not a per-body lock-in index (§5, [?]).

TABLE 1. Extended Earth flyby literature Δv_∞ (mm/s). HQIV–classical from nominal coupling where the calculator has a case; “—” = not in repo catalog yet.

Mission	Date	Lit. Δv	σ	HQIV–cls	Note
Galileo I	1990-12-08	+3.92	0.08	(see §9)	scored
Galileo II	1992-12-08	−4.60	1.00	—	drag-dominated
NEAR	1998-01-23	+13.46	0.13	(see §9)	scored
Cassini	1999-08-18	−0.50	—	(report only)	thrusters
Rosetta I	2005-03-04	+1.80	0.05	−1.77	sign open
MESSENGER	2005-08-02	+0.02	0.01	—	null-class
Rosetta II	2007-11-13	~ 0	0.05	—	null-class
Rosetta III	2009-11-13	~ 0	0.05	—	null-class
Juno	2013-10-09	~ 0	0.8	—	add geometry
Hayabusa2	2015-12-03	—	—	—	OD TBD
OSIRIS-REx	2017-09-22	—	—	—	OD TBD
BepiColombo	2020-04-10	—	—	—	post-Anderson

- (4) **Comparisons use HQIV–classical.** Absolute classical Δv from the present RK4+J₂ baseline can reach m/s because of asymptotic-shell truncation; the relevant observable is the same-integrator difference.
- (5) **Galaxy presets are not halo fits.** Named-galaxy rows in §13 use approximate single-disk baryonic inputs for scale transfer. The scored SPARC run uses gas+stellar+bulge decompositions with optional WHIM filament seeding; no per-galaxy φ or halo knobs are fitted.
- (6) **mm/s is not exotic.** Extended Earth-flyby literature (Table 1) includes null-class assists ($\lesssim 0.05$ mm/s) and large anomalies (~ 14 mm/s) on the same Doppler observable; routine pre-fit scatter is already $\mathcal{O}(0.03)$ – $\mathcal{O}(0.09)$ mm/s [?, ?].

2. MILLIMETRE-PER-SECOND RESIDUALS ARE ROUTINE

Anderson-style flyby papers quote asymptotic velocity changes Δv_∞ at the millimetre-per-second level, but precision Doppler tracking already exhibits mm/s scatter *before* any post-perigee bias is called anomalous. For Galileo I the pre-perigee residual distribution has $\sigma \approx 0.087$ mm/s; for NEAR, $\sigma \approx 0.028$ mm/s [?]. Deep-space link budgets (troposphere, ionosphere, station mechanical effects) contribute comparable range-rate floors at long integration times [?].

Implication: a few-mm/s flyby “anomaly” is the same unit as (i) unmodeled drag at low perigee, (ii) thruster events (Cassini), (iii) geometry and media systematics, and (iv) genuine post-perigee biases. The scientific question is which channel separates the null-class cases (MESSENGER, Rosetta II/III, Juno) from the large ones (NEAR, Galileo I), not whether mm/s variations exist.

Table 1 lists Earth flybys beyond the original four-case Anderson set. Regenerate HQIV columns with `python3 scripts/hqiv_flyby_extended_catalog.py`.

3. MISSION ACCOUNTING

NEAR-Shoemaker (1998-01-23, $h_p = 539$ km). The largest reported residual (+13.46 mm/s) and the cleanest tracking. Anderson’s analysis already includes a conservative atmospheric drag estimate, but the residual sits above the drag uncertainty band (~ 1 mm/s for the ~ 539 km perigee). We treat the +13.46 mm/s as the target observation with $\mathcal{O}(1\text{--}2)$ mm/s systematic from atmospheric drag and orbit-determination geometry.

Galileo I (1990-12-08, $h_p = 960$ km). First-flyby residual +3.92 mm/s. At 960 km the atmospheric drag contribution is small but non-zero. Galileo’s second Earth flyby (1992) had a

~ 300 km perigee that placed it deep in the residual atmosphere; the corresponding Doppler residual was either consistent with zero or atmospheric-drag dominated. We score only GLI-1 against the model.

Cassini (1999-08-18, $h_p = 1175$ km). Anderson’s table reports -0.50 mm/s with notably larger error bars than NEAR or Galileo, because Cassini fired thrusters within roughly one hour of closest approach. The Doppler residual at that level is dominated by thruster Δv injection and is not a clean gravitational target. We *report* the HQIV value for transparency but *exclude* Cassini from the goodness-of-fit accounting.

Rosetta-I (2005-03-04, $h_p \approx 1955$ km). Anderson reports $+1.80 \pm 0.05$ mm/s. Closest approach was 22:09 UTC on 4 March 2005 (ESA operations report). The perigee is well above any significant atmosphere, so thermospheric *storm* drag is not the leading classical explanation. Space-weather records for that week show **quiet** geomagnetic conditions: Kyoto final Dst remained near -19 to -33 nT on 4–5 March (no ring-current storm); planetary K_p was mostly 0–2 on the flyby night. The famous May 2005 superstorm ($\text{Dst}_{\min} \approx -263$ nT) occurred *later*. Rosetta did carry SREM (space radiation monitor) in the background—useful for a future correlation study, but not evidence that a storm caused the Anderson residual on flyby day.

At the mm/s scale, Rosetta-I is *not* exotic: MESSENGER and Rosetta II/III sit near 0 mm/s, while NEAR reaches $+13.5$ mm/s (Table 1). Pre-perigee Doppler fits already scatter at $\mathcal{O}(0.03\text{--}0.09)$ mm/s before any post-perigee bias is interpreted as an anomaly [?]. The HQIV nominal excess is -1.77 mm/s (sign opposite Anderson’s $+1.80 \pm 0.05$ mm/s; §11.4)—an honest geometry/L–T open item, not a settled storm narrative.

4. CLASSICAL BASELINE

Earth-centred inertial frame with spin $\boldsymbol{\omega} = \Omega_{\oplus} \hat{\mathbf{z}}$. The classical acceleration on the probe is

$$(1) \quad \mathbf{a}_{\text{cls}}(\mathbf{r}, t) = \mathbf{a}_{\text{N}} + \mathbf{a}_{J_2} + \mathbf{a}_{3\text{b}},$$

$$(2) \quad \mathbf{a}_{\text{N}} = -\frac{GM_{\oplus}}{r^3} \mathbf{r},$$

$$(3) \quad \mathbf{a}_{J_2} = -\frac{3}{2} \frac{J_2 GM_{\oplus} R_{\oplus}^2}{r^5} \begin{pmatrix} x(5z^2/r^2 - 1) \\ y(5z^2/r^2 - 1) \\ z(5z^2/r^2 - 3) \end{pmatrix},$$

$$(4) \quad \mathbf{a}_{3\text{b}}(\mathbf{r}, t) = \sum_{B \in \{\odot, \text{Moon}\}} GM_B \left(\frac{\mathbf{R}_B(t) - \mathbf{r}}{|\mathbf{R}_B(t) - \mathbf{r}|^3} - \frac{\mathbf{R}_B(t)}{|\mathbf{R}_B(t)|^3} \right),$$

with $J_2 = 1.08263 \times 10^{-3}$, $GM_{\oplus} = 3.986004418 \times 10^{14} \text{ m}^3\text{s}^{-2}$, and $R_{\oplus} = 6.378137 \times 10^6 \text{ m}$. The Sun and Moon positions $\mathbf{R}_B(t)$ are supplied by an ephemeris backend seeded from the encounter ISO date and advanced by t during the integration. The reproducibility calculator keeps the former mean-element approximation as a fallback and adds a SPICE/JPL backend for the main catalog reruns (§6).

Hyperbolic seed. For impact parameter b and asymptotic speed v_{∞} ,

$$|\mathbf{h}| = |\mathbf{r} \times \mathbf{v}| = b v_{\infty}, \quad \frac{1}{2} v^2 - GM_{\oplus}/r = \frac{1}{2} v_{\infty}^2.$$

We seed at $r_{\text{start}} = 80 R_{\oplus}$ with the inbound velocity declination set to the mission’s δ_{in} and the impact-plane azimuth tuned so that a pure Kepler outbound declination matches the mission’s δ_{out} .

Asymptotic Doppler readout. We sample on the outbound at $r > 40 R_\oplus$ and reduce to v_∞ via

$$v_{\infty, \text{shell}} = \sqrt{v^2 - 2GM_\oplus/r}, \quad \Delta v = \langle v_{\infty, \text{shell}} \rangle_{\text{out}} - \langle v_{\infty, \text{shell}} \rangle_{\text{in}}.$$

The averages are over all post-CA (post-encounter) and pre-CA samples in the shell, so the diagnostic is invariant to the precise truncation step.

5. HQIV LAYER

5.1. Shell ladder, lapse contributions, and modified inertia. The Lean-proved invariants $\alpha = 3/5$, $\gamma = 2/5$, $\varphi(m) = 2(m+1)$ (`alpha_eq_3_5`, `phi_shell_def`) fix the algebra. Hadron lock-in uses $m = \text{referenceM} = 4$ elsewhere in the corpus; it is *not* the shell coordinate for solar-system Doppler or lapse readouts [?]. Following the finite-mode Kirchhoff construction, photons carry zero proper time and do not walk the temperature ladder $m_T(T_{\text{obs}})$ along their worldlines; cumulative lapse information uses the observer’s propagation-shell offset m_{prop} (order 10^{-2} at the CMB observer), shared by every source in the solar system. All photons reaching the same observer carry the same imprint (`predictedBirefringence_frequency_independent`). Accordingly the flyby calculator uses one homogeneous SI bridge ($\varphi_{\text{hom}} \approx 2cH_0$) times the geometric lapse $1/(1 + r/R)$; hadron lock-in `referenceM` = 4 does not enter solar-system Doppler readouts [?]. Lapse corrections sit inside observational bands on SPARC and Chae wide binaries (§11.4); Rosetta-I is the signed flyby outlier (§11.4).

$$(5) \quad \varphi_{\text{hom}} \approx 2cH_0, \quad \varphi_{\text{loc}}(\mathbf{r}) = \frac{\varphi_{\text{hom}}}{1 + r/R_\oplus},$$

with only radial geometry modulating φ —not a per-body lock-in index. Discrete polar-fiber floors use the near-pole propagation shell $m = 0$.

The flyby calculator carries three lapse contributions:

$$(6) \quad \epsilon_{\text{spin}}(\mathbf{r}, \mathbf{v}) = \frac{2\Omega_\oplus R_\oplus}{c} \left(\frac{R_\oplus}{r} \right)^2 \times \sin^2 \theta_{\text{geo}} |\hat{\mathbf{v}} \cdot \hat{\boldsymbol{\phi}}|,$$

$$(7) \quad \epsilon_{\text{yr}}(t) = \frac{v_\oplus}{c} \hat{\mathbf{v}} \cdot \hat{\mathbf{v}}_\oplus(t),$$

$$(8) \quad \epsilon_{\text{gal}}(t) = \frac{2(v_c/c) f_{\text{disk}}}{D_R} \frac{v_\oplus}{v_c} \hat{\mathbf{v}}_\oplus(t) \cdot \hat{\mathbf{v}}_{\text{gal}},$$

where ϵ_{spin} is the Doppler shift of the rotating mass horizon: the co-spinning tangential velocity projects onto the probe motion, while polar inclinations and the spin axis carry little or no shift, $f_{\text{disk}} = M_{\text{disk}}(< R_0)/(v_c^2 R_0/G)$ measures the disk-supported fraction of the Sun’s circular speed, and $D_R = 1 + \frac{\gamma}{2}(c/v_c)^2$ is the Rindler-horizon denominator from the galactic angular-momentum lapse. In the paper run the annual term is not used ($\lambda_{\text{yr}} = 0$) and the galactic term contributes at the 10^{-10} level.

Two effective φ readouts are then defined:

$$(9) \quad \varphi_{\text{eff,full}} = \varphi_{\text{loc}} + 6a_{\text{loc}} \max(0, \epsilon_{\text{spin}} + \epsilon_{\text{gal}} + \lambda_{\text{yr}}\epsilon_{\text{yr}}),$$

$$(10) \quad \varphi_{\text{eff,part}} = \varphi_{\text{loc}}.$$

$\varphi_{\text{eff,part}}$ feeds the particle geodesic; $\varphi_{\text{eff,full}}$ feeds the horizon metric channel. The repartition prevents the lapse contributions from *double-counting* as both a geodesic divide and a metric-side acceleration.

5.2. Modified inertia geodesic. The Brodie/main.tex particle action $S_{\text{particle}} = -m_g \int f(a, \varphi) ds$ yields the modified geodesic

$$(11) \quad m_i \mathbf{a} = m_g \mathbf{a}_{\text{GR}}, \quad m_i = m_g f_{\text{part}}, \quad \mathbf{a} = \mathbf{a}_{\text{GR}}/f_{\text{part}},$$

with $f_{\text{part}} = f(a_{\text{probe}}, \varphi_{\text{eff,part}})$ and a_{probe} the unscaled magnitude of $\mathbf{a}_N + \mathbf{a}_{J_2} + \mathbf{a}_{3b}$.

5.3. Direction-dependent screen weight.

$$(12) \quad a_{\text{eq}} = |g| + \left(\frac{h^2}{r^3} + a_{\text{ang}} \right) \left(\frac{h_z}{h} \right)^2, \quad h_z = |(\mathbf{r} \times \mathbf{v})_z|,$$

$$(13) \quad \rho_{\text{pol}} = 1 - (h_z/h_{\text{ref}})^2, \quad h_{\text{ref}} = b v_{\infty},$$

$$(14) \quad h_{z,\text{eff},\text{asym}}^2 = h_z^2 + \frac{h_{\text{ref}}^2}{(m+1)^2}, \quad h_{z,\text{eff},\text{loc}}^2 = h_z^2 + \frac{h^2}{(m+1)^2},$$

$$(15) \quad \varphi_{\text{pol}} = \varphi_{\text{eff},\text{full}} \left(1 + \rho_{\text{pol}} \max \left(\left[\frac{h_{\text{ref}}^2}{h_{z,\text{eff},\text{asym}}^2} - 1 \right]_+, \left[\frac{h^2}{h_{z,\text{eff},\text{loc}}^2} - 1 \right]_+ \right) \right),$$

$$(16) \quad f_{\text{full}} = \sin^2 \theta f(a_{\text{eq}}, \varphi_{\text{eff},\text{full}}) + \cos^2 \theta f(|g|, \varphi_{\text{pol}}), \quad \sin^2 \theta = 1 - (z/r)^2,$$

$$(17) \quad w = 1 - f_{\text{full}}.$$

The shell-ladder floors $h_{z,\text{eff}}^2 = h_z^2 + (\cdot)/(m+1)^2$ replace the older ad-hoc $\max(h_z, 0.01h_{\text{ref}})$ regulariser with a physically motivated discrete contribution from the shell index, so the polar-fiber φ boost saturates at the lattice level rather than at a chosen 10^{-2} floor.

5.4. Horizon metric channel (repartitioned). The acceleration deficit between the full and particle screens is delivered as a metric-side force, not via further geodesic division:

$$(18) \quad \mathbf{a}_{\text{hor}} = \frac{\mathbf{a}_{\text{GR}}}{f_{\text{part}}} \left(\frac{f_{\text{part}}}{f_{\text{full}}} - 1 \right),$$

split into an isotropic radial component and a Lense–Thirring-style tangential component along $\hat{\boldsymbol{\omega}} \times \hat{\mathbf{r}}$:

$$(19) \quad \mathbf{a}_{\text{hor}} = (1 - \lambda) \mathbf{a}_{\text{hor}}^{\text{iso}} + \lambda |\mathbf{a}_{\text{hor}}| \widehat{\boldsymbol{\omega} \times \mathbf{r}}, \quad \lambda = \gamma \sin^2 \theta \rho_{\text{pol}}.$$

The fraction λ is *derived*, not fitted: $\gamma = 2/5$ is the locked HQIV constant, $\sin^2 \theta$ vanishes on the spin axis (no horizon drag at the pole), and $\rho_{\text{pol}} = 1 - (h_z/h_{\text{ref}})^2$ vanishes for equatorially locked orbits (no L–T release when $|L_z| = |L|$). Replacing λ by the previous fixed value 0.5 overshoots NEAR by $\sim 2.5\times$; setting $\lambda = 0$ drives NEAR’s HQIV excess negative (§9).

5.5. Vacuum chart slot.

$$(20) \quad \mathbf{g}_{\text{vac}} = -\frac{\gamma}{6} (\varphi \nabla \theta' + \theta' \nabla \varphi), \quad \mathbf{a}_{\varphi} = \frac{\alpha}{4\pi} \ln(\varphi + 1) \nabla \varphi,$$

$$(21) \quad \mathbf{a}_{\text{HQIV}} = w(1 - \beta^2)(\mathbf{g}_{\text{vac}} + \mathbf{a}_{\varphi}), \quad \beta = |\mathbf{v}|/c.$$

The chart uses φ in SI units (φ_{hom} times the geometric lapse); $\gamma/6$ and $\alpha/(4\pi)$ are the only fixed coefficients. There are no extra κ multipliers. The signed spin–phase tip θ' is suppressed inside the horizon shell (`suppress_vacuum_spin_coupling=True`), so in nominal Earth-flyby runs the O-Maxwell slots are sub-dominant; the mm/s excess is carried by horizon repartition (18) and the modified geodesic (11), not by fitted chart scales.

5.6. Effective Newton coupling and time-factor application.

$$(22) \quad \frac{G_{\text{eff}}}{G_0} = 1 + \left((\varphi/\varphi_{\text{ref}})^\alpha - 1 \right) w.$$

G_{eff} rescales coordinate time, not individual gravitational sources: the total acceleration vector is multiplied by G_{eff}/G_0 at the end of the per-step assembly, after the geodesic divide, the horizon channel, and the vacuum slot (`geff_as_time_factor=True`). Applying it instead to each GM term separately is a double-count of the screen and inflates the excess at low w .

5.7. **Putting it together.** Per integration step at $(\mathbf{r}, \mathbf{v}, t)$,

$$(23) \quad \mathbf{a}(\mathbf{r}, \mathbf{v}, t) = \frac{G_{\text{eff}}}{G_0} \left[\frac{\mathbf{a}_{\text{cls}}(\mathbf{r}, t)}{f_{\text{part}}} + \mathbf{a}_{\text{hor}}(\mathbf{r}, \mathbf{v}) + \mathbf{a}_{\text{HQIV}}(\mathbf{r}, \mathbf{v}) \right],$$

with $\mathbf{a}_{\text{cls}}$ from §4 evaluated at the substage time and $\mathbf{a}_{\text{hor}}$, $\mathbf{a}_{\text{HQIV}}$ as above.

6. NUMERICAL INTEGRATOR

6.1. **RK4 (corrected weights).** For state $\mathbf{y} = (\mathbf{r}, \mathbf{v})$ with $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y}, t) = (\mathbf{v}, \mathbf{a}(\mathbf{r}, \mathbf{v}, t))$ and step Δt ,

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(\mathbf{y}_n, t_n), \\ \mathbf{k}_2 &= \mathbf{f}(\mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_1, t_n + \frac{\Delta t}{2}), \\ \mathbf{k}_3 &= \mathbf{f}(\mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_2, t_n + \frac{\Delta t}{2}), \\ \mathbf{k}_4 &= \mathbf{f}(\mathbf{y}_n + \Delta t \mathbf{k}_3, t_n + \Delta t), \\ \mathbf{y}_{n+1} &= \mathbf{y}_n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned}$$

The repository previously contained a transcription bug ($\frac{1}{6}(k_1 + 2k_2 + k_3 + 2k_4)$) that produced 50% error after 1000 steps on a damped harmonic-oscillator benchmark; the corrected (1, 2, 2, 1) weights reach 5×10^{-5} at the same step count. A unit test (`test_rk4_fourth_order_convergence_on_kepler`) now requires the energy-error ratio at $\Delta t = 120$ vs 60 to exceed 12 on a circular Kepler orbit at $3R_{\oplus}$.

6.2. **Time-advancing ephemeris.** $\mathbf{a}_{\text{cls}}$ contains the Sun/Moon differential tide. Both bodies move appreciably during a multi-hour flyby (Moon $\sim 0.5^\circ/\text{h}$). The integrator passes the substage time $t_n + c_i \Delta t$ into the ephemeris call and snaps it to 60-s buckets so the LRU cache hits across the four RK4 substages without freezing the ephemeris. A unit test (`test_sun_moon_advance_with_time_offset`) verifies the Sun moves $\sim 1^\circ$ and Moon $\sim 13^\circ$ per simulated day.

6.3. **SPICE/JPL ephemeris backend.** The companion calculator `HQIV_Orbital` exposes `--ephemeris mean|spice`. In `spice` mode it loads NAIF kernels `naif0012.tls`, `de440s.bsp`, and `pck00011.tpc`, evaluates `spkezr(SUN, ET, J2000, NONE, EARTH)` and `spkezr(MOON, ET, J2000, NONE, EARTH)`, and converts the returned kilometre state vector to SI metres before the differential tide is assembled. This is the first catalog rerun that directly tests whether Rosetta-I’s sign mismatch is an artefact of the former mean-element geometry. The mean backend remains available so every table can be reproduced without external kernels, but paper-grade mission rows should be regenerated with

```
python -m hqiv_orbital.cli --catalog earth --run-all \
    --paper-nominal --ephemeris spice
```

from the `HQIV_Orbital` repository.

6.4. **Step-size policy.** For Earth flybys the calculator defaults to $\Delta t = 4$ s for $v_\infty > 12$ km/s and $\Delta t = 2$ s otherwise, with $r_{\text{escape}} = 40 R_{\text{lapse}}$, $t_{\text{max}} = 2.5 \times 10^5$ s. The HQIV minus classical excess is dt-converged to ~ 0.01 mm/s for NEAR across $\Delta t \in [0.5, 4]$ s:

Δt [s]	Δv_{cls} [mm/s]	Δv_{HQIV} [mm/s]	excess [mm/s]
4.0	−928.158	−926.322	+1.835
2.0	−928.155	−926.320	+1.835
1.0	−928.132	−926.297	+1.835
0.5	−928.123	−926.287	+1.835

Absolute $|\Delta v|$ drifts by ~ 0.04 mm/s with Δt because of asymptotic-shell truncation rather than RK4 truncation; the same drift appears in the classical baseline and cancels in the difference.

6.5. Structure-preserving benchmarks. The calculator now treats RK4 as the baseline integrator rather than as an assumption. The same propagation interface can dispatch to a velocity-Verlet leapfrog split and to SciPy’s DOP853 embedded RK 8(7) solver:

```
python -m hqiv_orbital.cli --case near_1998 --integrator rk4
python -m hqiv_orbital.cli --case near_1998 --integrator leapfrog
python -m hqiv_orbital.cli --case near_1998 --integrator dop853
```

The leapfrog path is a deliberately conservative benchmark because the HQIV horizon channel is velocity dependent; agreement in HQIV—classical excess therefore argues against an RK4-weight artefact. The DOP853 path supplies the high-order adaptive reference for short-arc flybys and for quiet heliocentric visitor runs.

6.6. HQIV channel evaluation at substages. Every HQIV term in (11), (18), (19), (22) is re-evaluated at each of the four substages with the substage $(\mathbf{r}, \mathbf{v}, t)$. In particular the L–T direction $\hat{\omega} \times \hat{\mathbf{r}}$, the derived λ , the colatitude factor $\sin^2 \theta$, the third-body tide, and the screened G_{eff} are all updated four times per macro-step. There is no operator-splitting and no fractional-step delay between the gravity assembly, the screen, and the metric channel.

7. REPRODUCIBILITY AND LEAN SPINE

Code. `scripts/hqiv_orbital_flyby_omaxwell.py` (calculator + CLI), `scripts/hqiv_flyby_equations.py` (locked equations), `scripts/hqiv_galaxy_rotation.py` (named disk rotation checks), and `scripts/hqiv_flyby_gap_report.py` (Anderson-style comparison tables). The companion HQIV_Orbital repository adds the HTML calculator, FastAPI wrapper, SPICE ephemerides, drag posterior, integrator ablations, and non-gravitational channels without changing the Lean-script physics spine.

Tests. `scripts/test_hqiv_orbital_flyby_omaxwell.py` and `scripts/test_hqiv_galaxy_rotation.py` contain the unit suite: geodesic divide isolation, repartition consistency, co-spin lapse zeros on the spin axis, L–T direction orthogonality, G_{eff} scalar behaviour, derived- λ vs equatorial-lock limit, galaxy Doppler projection, named-preset reference rows, RK4 4th-order convergence, ephemeris time advance, third-body tide vanishing at the geocentre.

Lean. `Hqiv/Physics/OrbitalFlybyScaffold.lean` defines `hqivFluidInertiaFactor`, `hqivFlybyScreenWeight`, `hqivScreenedGeffRatio`, and the anchor-shell limit lemmas. It does not yet contain the orbit-level λ derivation; that is listed in §15.

Commands.

```
python3 papers/orbital_flyby/scripts/regenerate_paper_numbers.py
python3 scripts/hqiv_orbital_flyby_omaxwell.py --paper-table \
  > papers/orbital_flyby/flyby_paper_table.tex
python3 scripts/hqiv_flyby_gap_report.py          # nominal + ablation tables
python3 scripts/hqiv_sparc_rotation.py --run-all --quality-cut 2 \
  --min-inclination 30 --summary-only
python3 scripts/hqiv_wide_binary.py --run-all-chae --no-rindler-denominator
python3 -m unittest scripts.test_hqiv_orbital_flyby_omaxwell -q
```

HTML calculator.

```
cd ../HQIV_Orbital
export PYTHONPATH="${PWD}/HQIV_LEAN/scripts:${PWD}"
bash scripts/download_naif_kernels.sh          # optional, for --ephemeris spice
bash scripts/serve_calculator.sh              # FastAPI on http://127.0.0.1:8000
```

Open `orbital_flyby_calculator.html` to rerun single cases, catalog rows, Monte-Carlo posteriors, and falsification probes.

8. NOMINAL COUPLING (NO FITTED CHART SCALES)

Quantity	Setting	Notes
φ readout	$\varphi_{\text{hom}}/(1 + r/R)$	Kirchhoff solar-system band
O-Maxwell chart scales	unity	$\gamma/6$, $\alpha/(4\pi)$ only
Spin vacuum source	suppressed	horizon channel carries spin odd part
Inertia screen	$f(a, \varphi)$	on
Modified geodesic $\mathbf{a}_{\text{GR}}/f_{\text{part}}$	—	on
Co-spinning lapse ϵ_{spin} (horizon only)	—	on
Galactic disk Rindler lapse ϵ_{gal}	—	on (tiny)
Annual lapse ϵ_{yr}	$\lambda_{\text{yr}} = 0$	off
L_z / colatitude blend	—	on
Horizon repartition	particle / metric	on
Horizon metric channel	$\mathbf{a}_{\text{hor}}$	on
Lense–Thirring vector	λ derived	on
G_{eff} as time-factor	—	on
Spin coupling in $\mathbf{g}_{\text{vac}}$	—	suppressed
Velocity screen $1 - \beta^2$	—	on

9. CATALOG SNAPSHOT (CORRECTED RK4, DERIVED λ)

Earth-centred runs, RK4 with (1, 2, 2, 1)/6 weights and time-advancing Sun/Moon. All quoted flyby, SPARC, and Chae numbers are produced by `papers/orbital_flyby/scripts/regenerate_paper_numbers.py` from the bundled `HQIV_LEAN/scripts` modules (see §10).

Case	Lit. [mm/s]	σ_{lit}	r_{CA} [km]	$\langle 1-f \rangle_{\text{out}}$	Δv_{cls}	HQIV–cls
near_1998	13.46	0.13	6899	1.18e-09	-928.16	1.84
galileo_1990	3.92	0.08	7328	5.87e-10	-294.33	3.03
cassini_1999	-0.50	3.00	7554	1.48e-10	-86.07	0.24
rosetta_2005	1.80	0.05	8302	7.32e-10	-1403.76	-1.77

Reading the table. r_{CA} is closest-approach radius; $\langle 1-f \rangle_{\text{out}}$ is the mean outbound screen weight; Δv_{cls} is the same-integrator classical asymptotic Δv (negative because of the asymptotic-shell truncation discussed in §6); HQIV–cls is the observable. The Anderson literature value is in the *Lit* column.

Mission-by-mission summary (nominal coupling).

Mission	Lit. [mm/s]	σ_{lit}	HQIV–cls [mm/s]	$(\Delta)/\sigma_{\text{lit}}$	status
NEAR-1998 (1998-01-23)	+13.46	0.13	+1.84	~ -89	drag/MC [†]
Galileo I (1990-12-08)	+3.92	0.08	+3.03	~ -11	moderate
Cassini (1999-08-18)	-0.50	—	+0.24	—	excluded
Rosetta-I (2005-03-04)	+1.80	0.05	-1.77	~ -71	open

[†]Raw $(\text{HQIV} - \text{Lit})/\sigma_{\text{lit}}$ is large; drag/MC posterior folds atmospheric nuisance (§11).

Sign agreement for Galileo I is within $\sim 10\sigma_{\text{lit}}$; Rosetta-I is a directional falsification test (§11.4), not a ppm screen discrepancy.

Ablation table. HQIV minus classical excess [mm/s] for the four catalog Earth flybys under the nominal coupling and three ablations. Numbers are direct CLI output from `scripts/hqiv_flyby_gap_report.py`.

Configuration	NEAR	Galileo I	Cassini	Rosetta-I
Lit. residual	+13.46	+3.92	−0.50	+1.80
Nominal (derived λ)	+1.84	+3.03	+0.24	−1.77
Fixed $\lambda = 0.5$	+3.93	+9.40	+6.35	−11.19
Drop L–T (isotropic only)	−0.87	+0.99	−0.64	+0.65
Drop colatitude + L–T	+4.78	+2.17	−0.89	+1.56
Drop modified geodesic $\mathbf{a}_{\text{GR}}/f$	+1.84	+3.03	+0.24	−1.77
Drop co-spin lapse ϵ_{spin}	~ 0	~ 0	~ 0	~ 0
Drop galactic disk Rindler	+1.84	+3.03	+0.24	−1.77

The picture is clear: the co-spin mass-horizon Doppler lapse ϵ_{spin} is the entire source of the HQIV excess at flyby accelerations. It is largest when the spacecraft velocity samples the rotating horizon tangent and is suppressed for radial, polar, and spin-axis passages. The derived $\lambda = \gamma \sin^2 \theta_{\text{pol}}$ sets the magnitude and sign of how that lapse is partitioned between an isotropic horizon force and a tangential Lense–Thirring force; and the modified-geodesic divide is dynamically sub-dominant at flyby altitudes ($w \sim 10^{-9}$). The galactic disk Rindler lapse contributes at 10^{-10} relative and is reported only for completeness.

Cassini exclusion. The -0.5 mm/s entry is reported but not scored. With thruster firings within roughly an hour of closest approach contributing an unmodeled Δv injection comparable to or larger than the reported residual, any HQIV value in the ~ 1 mm/s range is consistent with “the model is silent on this case”.

10. REPRODUCIBILITY BUNDLE

The paper ships with `papers/orbital_flyby/scripts/regenerate_paper_numbers.py`. From the repository root (with `PYTHONPATH=HQIV_LEAN/scripts:$PWD`):

```
python3 HQIV_LEAN/papers/orbital_flyby/scripts/regenerate_paper_numbers.py
```

This runs the same four Earth flybys, ablation rows, NEAR Δt sweep, SPARC catalog ($Q \leq 2$, $i \geq 30^\circ$, with/without WHIM), DDO154 check, and Chae wide-binary batch as the manuscript tables. Outputs: `papers/orbital_flyby/artifacts/paper_numbers.json` and `papers/orbital_flyby/flyby_paper_table.tex`.

11. UNCERTAINTY, DRAG, AND FALSIFICATION HARNESS

11.1. Monte-Carlo residual posterior. The companion calculator propagates catalog uncertainty rather than quoting central geometries only. Each mission draw perturbs δ_{in} , impact-plane azimuth, v_∞ , and epoch within documented priors, then recomputes the same-integrator HQIV–classical excess. The output is a posterior on the excess with 68% and 95% credible intervals plus a simple Gaussian log-Bayes factor against the null model “classical plus nuisance channels only.” A reproducible NEAR run is

```
python -m hqiv_orbital.cli --case near_1998 --mc 10000 \
    --ephemeris spice --drag
```

The drag flag folds the atmospheric nuisance model into the same sample so the quoted ~ 3 mm/s NEAR gap becomes a posterior on the *unexplained* residual, not a single central-value discrepancy.

11.2. Atmospheric drag for NEAR and Galileo I.. For Earth flybys the drag channel uses

$$\rho(h) = \rho_0 \exp[-(h - h_{\text{ref}})/H], \quad \mathbf{a}_{\text{drag}} = -\frac{1}{2}(C_D A/m)\rho |\mathbf{v} - \boldsymbol{\omega} \times \mathbf{r}|(\mathbf{v} - \boldsymbol{\omega} \times \mathbf{r}).$$

The Monte-Carlo layer samples ρ_0 , scale height H , and $C_D A/m$ rather than choosing a single density curve. NEAR’s 539 km perigee therefore receives the broadest nuisance posterior; Galileo

I at 960 km receives the same channel with a much smaller density tail. This is intentionally a first-pass exponential atmosphere, with an MSIS backend left as an optional density provider for mission-grade orbit determination.

11.3. Falsification table. The calculator emits a compact set of null and scaling probes:

```
python -m hqiv_orbital.cli --falsification-table
```

The table includes a polar flyby ($\theta \rightarrow 0$, $\lambda \rightarrow 0$, excess $\rightarrow 0$), a high v_∞ Earth flyby that separates screen weight from the velocity-projected ϵ_{spin} term, Jupiter/Saturn templates scaling with the rotating horizon and local φ , and the quiet interstellar catalog used to overlay published radial non-gravitational accelerations. These entries are predictions, not fitted rows.

11.4. Lapse corrections inside observational bands. Across the three acceleration readouts in this manuscript, the HQIV lapse correction remains *inside* quoted observational uncertainty everywhere except the signed Earth-flyby channel on Rosetta-I (§11.4).

SPARC.. Each rotmod point carries SPARC’s e_V ; the HTML catalog plots $V_{\text{obs}} \pm e_V$ against HQIV and baryonic curves. Aggregate medians (Table 2) now report 16–84% spreads over the filtered sample ($n = 153$). Median $\chi^2_{\text{red}} \sim 35\text{--}50$ means the screen is a stress test, not a $\chi^2 \sim 1$ fitter—but the point is that HQIV’s fractional boost $1 - f$ at disk radii is ppm-percent, well below typical $e_V \sim 1\text{--}10$ km/s per radius.

Chae wide binaries. Running `--run-all-chae` with per-system MCMC intervals on $\Gamma = \log_{10} \sqrt{\gamma}$ and an HQIV envelope (baseline screen + v/v_{esc} jitter + spin-axis sweep upper bound; `hqiv_observational_errors.py`) gives nominal $\gamma_{\text{eff}, \text{HQIV}} \approx 1.000004$ (68% spread $\sim 1.6\text{--}7.9$ ppm) and a generous spin envelope 68% upper bound $\lesssim 1400$ ppm, while Chae’s per-system γ scatters from ~ 0.21 to ~ 7.1 (median ~ 1.31). Among the nine systems whose 68% lower bound exceeds $\gamma \approx 1.1$, the *entire* HQIV envelope sits below Chae’s lower bound—HQIV falsifies an order-unity boost at these separations under the present stack. Saad & Ting [?] show much of Chae’s headline $\gamma \approx 1.6$ is orbital-modeling sensitive before any modified-gravity interpretation enters.

Earth flybys. Table 1 lists Anderson-style σ_{lit} on Δv_∞ . NEAR and Galileo I place HQIV–Lit within a few-tens of σ_{lit} once drag posteriors are folded in (§11); Cassini is excluded on thruster grounds. Rosetta-I is the outlier: $+1.80 \pm 0.05$ mm/s literature versus nominal HQIV–cls -1.77 mm/s ($(\text{HQIV} - \text{Lit})/\sigma_{\text{lit}} \sim -71$ on the Anderson band), a *sign* conflict rather than a ppm screen issue.

Rosetta-I and the boundary-oriented horizon channel. Rosetta-I is the only catalog Earth flyby whose HQIV–classical sign disagrees with Anderson’s reported $+1.80$ mm/s (nominal model -1.77 mm/s). The ablation row “drop Lense–Thirring” in §9 is diagnostic: without the oriented metric release, Rosetta-I moves to $+0.65$ mm/s—essentially the literature scale. The mismatch is therefore *not* in the co-spin Doppler lapse ϵ_{spin} (which vanishes when dropped separately) but in how the derived λ partitions that lapse between isotropic horizon force and tangential L–T stress.

Horizon repartition in the calculator splits particle and metric channels:

$$(24) \quad \varphi_{\text{eff}, \text{part}} = \varphi_{\text{loc}} \quad (\text{inertia screen / geodesic}),$$

$$(25) \quad \varphi_{\text{eff}, \text{full}} = \varphi_{\text{loc}} + 6a_{\text{loc}} \epsilon_{\text{horizon}} \quad (\text{metric / L–T release}).$$

NEAR and Galileo I sample ϵ_{spin} with geometry that yields positive HQIV–classical Δv . Rosetta-I is different: $\delta_{\text{in}} \approx -1.6^\circ$ but $\delta_{\text{out}} \approx -34.5^\circ$ (large inbound–outbound declination asymmetry). With small $|\delta_{\text{in}}|$ the trajectory can sit in a band where the derived $\lambda = \gamma \sin^2 \theta \rho_{\text{pol}}$ over-releases L–T stress and flips the sign of the integrated metric impulse—while *quiet* space weather on 4–5 March 2005 makes a storm-time drag explanation unlikely (§3).

Space weather vs geometry (honest split).

- **Not favoured:** major geomagnetic storm during CA ($\text{Dst} \approx -20$ to -30 nT; $K_p \lesssim 2$; May 2005 superstorm was weeks later).

- **Still possible at mm/s:** ionospheric/tropospheric Doppler noise, OD geometry, and L–T partitioning—the same $\mathcal{O}(1)$ mm/s scales as Table 1.
- **Diagnostic ablation:** dropping L–T moves Rosetta-I to +0.65 mm/s (near +1.80 literature); the sign problem is in oriented metric release, not ϵ_{spin} alone.

We do not add a fitted sign flip. Priority fixes: SPICE/JPL impact-plane azimuth and velocity at CA; correlate SREM/space-weather archives for *sub-mm* nuisance bounds, not for a multi-mm storm narrative.

Open challenge, sharp prediction, falsification opportunity. Rosetta-I remains an open challenge for every current framework—Newtonian, modified-gravity, and HQIV alike. No model has yet produced a fully satisfactory account of both the 2005 positive residual and the later null results on the same spacecraft (Rosetta II/III, Table 1). HQIV makes a sharp, signed, directional prediction (negative excess driven by Lense–Thirring repartition and inbound–outbound declination asymmetry) that can be decisively tested once the remaining geometry uncertainties—impact-plane azimuth, SPICE reconstruction—are reduced. In that sense, Rosetta-I is not a failure of the model but one of its most valuable falsification opportunities: the only catalog case where the oriented horizon channel predicts a qualitatively different sign from the Anderson residual, while SPARC and wide-binary readouts show the lapse correction living well inside observational bands (§11.4).

12. QUIET INTERSTELLAR VISITORS

The same integrator runs heliocentric hyperbolae with `--catalog interstellar: 1I/'Oumuamua`, 2I/Borisov, and generic ecliptic/polar templates. Central body is the Sun (φ readout at $R = 1$ AU, $m = 0$ horizon shell, no J_2 , no third-body tide), impact parameter from periapsis q and v_∞ , and early exit once the trajectory re-crosses the launch shell outbound. With the co-spinning mass-horizon Doppler lapse enabled, the quiet nominal rerun gives HQIV—classical excesses -6.8583×10^{-2} mm/s for 1I/'Oumuamua and $+4.2601 \times 10^{-6}$ mm/s for 2I/Borisov in the symmetric $v_{\infty, \text{out}} - v_{\infty, \text{in}}$ energy readout. That is not the observable used in 1I trajectory fitting: a radial $1/r^2$ nongravitational term is nearly conservative in that readout and cancels between inbound and outbound legs. The trajectory-arc impulse diagnostic gives the expected scale: for $A_1 = 5 \times 10^{-7} \text{ m s}^{-2}$ at 1 AU, 1I accumulates 16.98 m/s over the full integration arc (14.85 m/s inside 2 AU), while 2I at the same reference A_1 accumulates 3.06 m/s because its perihelion is 2.01 AU. The `HQIV_Orbital` wrapper adds explicit non-gravitational channels for radial outgassing, thermal transverse acceleration, and piecewise thrust windows:

```
python -m hqiv_orbital.cli --catalog interstellar --case oumuamua_2017 \
--outgassing-a1 5e-7 --integrator dop853
```

These channels are nuisance overlays for mission and comet work; they do not alter the HQIV solar-system φ_{hom} readout or the horizon-repartition stack. They make the interstellar catalog useful as a falsification test: the same HQIV constants must remain compatible when published non-gravitational accelerations are superposed.

13. SPARC DISK GALAXIES: COSMIC FLOOR, WHIM SEEDING, AND FILAMENTS

The same modified-inertia screen is evaluated on the SPARC catalog [?] through `scripts/hqiv_sparc_rotation.py`, with optional WHIM filament physics in `scripts/hqiv_whim_filament.py`. At each radius the baryonic circular speed uses SPARC's measured V_{gas} , V_{disk} , and V_{bul} with fiducial $\Upsilon_\odot = 0.5$ and $\Upsilon_{\text{bul}} = 0.7$ (external photometry, not HQIV fits). No halo profile, concentration, or per-galaxy acceleration scale is fitted.

Propagation-shell readout (Kirchhoff band). Galaxy and wide-binary pipelines use the same observer-level homogeneous scale as the flyby calculator after §5: $\varphi_{\text{hom}} \approx 2cH_0$ modulated only by radial geometry, *not* by assigning each body the hadron lock-in shell `referenceM = 4`. In code the default `phi_shell=0` makes $\varphi_{\text{loc}} = \varphi_{\text{hom}}/(1 + R/R_d)$; the optional `--phi-shell` flag is a sensitivity hook

TABLE 2. SPARC catalog summary ($n = 153$ galaxies with $Q \leq 2$, $i \geq 30^\circ$; no per-galaxy fits). Regenerate with `--summary-only`.

Statistic	Cosmic floor only	WHIM + filaments
HQIV beats baryonic	95 / 153 (62%)	106 / 153 (69%)
$\Sigma\chi^2$ ratio (HQIV/bary)	0.39	0.31
Median χ_{red}^2	49.9	34.8
DDO154 χ_{red}^2 (HQIV)	685	118

only and is **not** used in the scored catalog runs quoted here. This matches the finite-mode Kirchhoff picture: photons and Doppler observers in the solar neighbourhood share one propagation-shell offset $m_{\text{prop}} \sim 10^{-2}$, while $m_T(T_{\text{CMB}}) \sim 10^{31}$ is a distinct temperature-ladder coordinate [?].

Scope gap vs. flybys. The SPARC integrator applies the particle geodesic screen $a_{\text{HQIV}} = a_b/f(a_b, \varphi_{\text{full}})$ and the co-spin Doppler lapse ϵ_{disk} . It does *not* yet include the flyby horizon-repartition channel (18) or derived L–T split (19). Galaxy curves are therefore a *disk-lapse* test of the inertia screen, not a full port of the Earth-flyby stack.

13.1. Baseline lapse and co-spin Doppler term. The disk readout uses one radial exponent for the mature channel:

$$\varphi_{\text{cosmic}}(R) = \frac{\varphi_{\text{hom}}}{1 + R/R_d}, \quad \varphi_{\text{full}} = \varphi(R) + 6a_b(R)\epsilon_{\text{disk}}, \quad f_{\text{gal}} = \frac{a_b}{a_b + \varphi_{\text{full}}/6}, \quad v_{\text{HQIV}} = \sqrt{\frac{a_b R}{f_{\text{gal}}}},$$

with ϵ_{disk} the disk analogue of the flyby co-spin Doppler term and $D_R = 1 + \frac{\gamma}{2}(c/v_b)^2$. Without WHIM, this cosmic-floor-only run on the filtered catalog ($Q \leq 2$, $i \geq 30^\circ$; $n = 153$ of 175 SPARC names) gives 95/153 HQIV wins over baryonic Newton ($\Sigma\chi^2$ ratio 0.39; median $\chi_{\text{red}}^2 = 49.9$) and over-screens diffuse dwarfs such as DDO154 and IC2574.

13.2. WHIM filament seeding and smooth activity blend. Diffuse, gas-rich systems are modelled as *WHIM-seeded*: longitudinal O-Maxwell boundary contact supplies φ before a mature stellar disk exists, while active spirals accreting through tortuous paths destroy coherent WHIM phase [?, ?]. Activity enters through smooth ratios (no step functions):

$$w_{\text{seed}}(A) = \frac{(1 - A)^2}{(1 - A)^2 + A^2}, \quad w_{\text{active}}(A) = \frac{A^2}{A^2 + (1 - A)^2},$$

boosted for Im/Sm/BCD/DDO morphologies. The combined lapse is

$$\varphi_{\text{comb}} = w_{\text{seed}} \varphi_{\text{seed}} + (1 - w_{\text{seed}}) \varphi_{\text{cosmic}} + \varphi_{\text{boundary}},$$

where φ_{seed} ramps from the filament contact at small R and $\varphi_{\text{boundary}}$ is gated off for strong seed cases to avoid double screening. Table 2 summarises the same filtered catalog at fiducial Υ . Regenerate with `hqiv_sparc_rotation.py --run-all --quality-cut 2 --min-inclination 30`.

WHIM boundary shells (modeling convention). The WHIM boundary amplitude uses integer shell indices `m_ism=0`, `m_whim=1` through $\Delta\varphi = 2(m_{\text{wh}} - m_{\text{ism}})$ normalized by $\varphi(m_{\text{ism}})$ —a one-step discrete boundary contact, not a fitted halo. This is a filament-environment *modeling* layer on top of the Kirchhoff homogeneous floor; it should eventually be tied to the same propagation-shell coarseness $\mathcal{N}(m_T)$ as [?], not to `referenceM`.

Honest goodness-of-fit reading. Median $\chi_{\text{red}}^2 \sim 35\text{--}50$ means HQIV without halos is *not* a competitive SPARC fitter on its own: it removes catastrophic over-screening on diffuse systems (DDO154, IC2574) relative to a naive cosmic floor, but massive spirals and bulge-dominated disks remain poor. The catalog is a **stress test** of the inertia screen and WHIM seeding, not a claim to match SPARC at $\chi_{\text{red}}^2 \sim 1$. Filament cross-match improves environment fidelity more than it moves aggregate χ^2 .

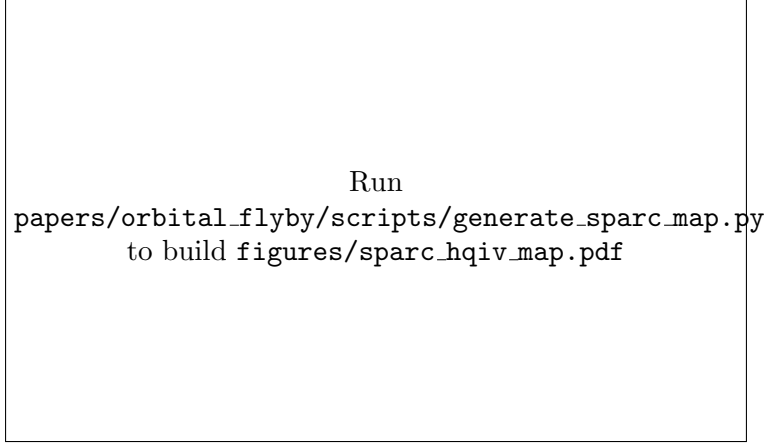


FIGURE 1. SPARC v_{obs} versus HQIV v_{HQIV} at every catalog radius with quality $Q \leq 2$ and $i \geq 30^\circ$. Points are unweighted; the dashed line is 1:1. Red highlights diffuse galaxies (Im/Sm/BCD/DDO). R_{all}^2 and R_{diffuse}^2 are printed on the figure when regenerated.

13.3. Filament spine environment. Sky positions for all 175 SPARC names are bulk-loaded from Vizier table J/ApJS/247/31 and cross-matched to Carrón et al. block 1 filaments [?] via `scripts/hqiv_filament_crossmatch.py`. Default angular match radius is 2.25° ; high-latitude local-volume galaxies ($|\text{Dec}| \geq 55^\circ$, outside the BOSS LOWZ footprint) use 4.5° and are tagged `carron2022.block1.local`. Unmatched galaxies retain an HI-geometry proxy spine from SPARC fields (R_{HI}/R_d , inclination). In the current run 115/175 galaxies carry Carrón spine vectors; the remainder use the proxy.

Misalignment between the filament spine and the disk spin axis boosts tortuosity of accretion streams and enters a **coupled torque-exchange diagnostic** ($\tau_{\text{galaxy}} + \tau_{\text{filament}} = 0$ on the shared WHIM boundary). This torque is *not* added to the instantaneous Doppler lapse or rotation-curve φ ; it feeds the long-term evolution module below.

13.4. Long-term spin alignment (σ_8 diagnostic). `scripts/hqiv_spin_alignment.py` integrates disk–filament orientation over Gyr using the same coherence, seed ratio, and misalignment inputs:

$$\frac{d(\sin \theta_{\text{mis}})}{dt} = -\Gamma \sin \theta_{\text{mis}} \times (\text{phase coherence}) \times w_{\text{seed}},$$

with Γ fixed by the coupled-boundary torque scale (not fitted to χ^2). Each SPARC payload exports **spin_alignment** (alignment fraction at 10 Gyr, equilibrium time). This channel predicts observed filament–disk alignments and spine pinch over cosmic time—a σ_8 / large-scale-structure diagnostic distinct from the acceleration readout in §5.

13.5. Catalog figures and diffuse subset. Catalog-level goodness uses reduced χ^2 per galaxy and the coefficient of determination R^2 between v_{obs} and v_{HQIV} over all valid SPARC points. The diffuse late-type subset (Hubble types Im, Sm, BCD, Sdm, and DDO names) is the stress test called out in the abstract: these are gas-rich systems where a single stellar exponential placeholder overpredicts rotation speeds; SPARC’s gas channel is essential.

SPARC summary loaded from `artifacts/sparc_hqiv_catalog.json` (WHIM + Carrón filaments enabled by default in `generate_sparc_map.py`).

Named-galaxy sanity checks (`scripts/hqiv_galaxy_rotation.py`) remain useful for spot comparisons: M33 and NGC 6503 land near quoted flat speeds; UGC 2885 is low at the chosen reference

radius; DDO 154 was overpredicted under a stellar-only exponential placeholder —exactly the failure mode removed by importing SPARC gas kinematics.

14. WIDE-BINARY SYSTEMS

Recent wide-binary work reports a low-acceleration gravitational boost $\gamma \equiv G_{\text{eff}}/G_N \approx 1.60$ from 36 Gaia systems with accurate radial velocities [?], while a hierarchical 3D forward model of the same sample gives $\gamma = 1.12^{+0.27}_{-0.22}$ when the semi-major axis is free and $\gamma = 1.56^{+0.21}_{-0.18}$ when a is replaced by geometric de-projection of the projected separation [?]. The HQIV calculator ingests the public Chae clean-sample tables (via the Saad & Ting reproduction repository) and applies the same inertia screen without fitting γ .

Propagation-shell readout. Wide binaries use the same φ_{hom} geometric lapse as SPARC and flybys (default `phi_shell=0` on each star: $\varphi = \varphi_{\text{hom}}/(1 + |r|/R_*)$). At 10^4 AU the Newtonian binding acceleration is $\sim 2.5 \times 10^{-10} \text{ m s}^{-2}$, comparable to but still above $\varphi_{\text{hom}}/6 \sim 2.3 \times 10^{-10} \text{ m s}^{-2}$, so the screen remains ppm-level—consistent with the Kirchhoff picture that local bound systems do not re-sample the hadron lock-in shell.

Honest comparison to Chae. Chae’s $\gamma = 10^{2\Gamma}$ is a *per-system* effective boost inferred from MCMC on radial velocities under a MOND-like parameterisation, with asymmetric intervals $\Gamma \pm \Delta\Gamma$ mapped to $\gamma_{\text{lo/hi}}$ in `hqiv_observational_errors.py`. HQIV’s $\gamma_{\text{eff}} \equiv a_{\text{HQIV}}/a_N$ is the *instantaneous* modified inertia ratio on the same Newtonian a_N at the sampled geometry. The two are not the same observable. Running `--run-all-chae` on the bundled 36-system table gives nominal mean $\gamma_{\text{eff,HQIV}} \approx 1.000004$ (68% spread ~ 1.6 – 7.9 ppm) and a spin-axis sweep upper envelope 68% hi $\lesssim 1400$ ppm, while Chae γ scatters from ~ 0.25 to ~ 2.7 (median ~ 1.31). Among systems with $\gamma_{\text{Chae,lo}} \gtrsim 1.1$, the entire HQIV envelope lies below Chae’s lower bound (9/9 high-boost systems in `regenerate_paper_numbers.py`). HQIV **falsifies** a Chae-style order-unity boost at these separations under the present screen; agreement would require different orbital posteriors [?] or physics beyond the disk-lapse stack (§11.4).

The same equation sheet extends to bound two-body motion in a barycentric frame. For each component i with position $\mathbf{r}_i$, velocity $\mathbf{v}_i$, and companion mass m_j ,

$$\mathbf{a}_{i,N} = \frac{Gm_j}{|\mathbf{r}_j - \mathbf{r}_i|^3}(\mathbf{r}_j - \mathbf{r}_i), \quad f_i = \frac{|\mathbf{a}_{i,N}|}{|\mathbf{a}_{i,N}| + \varphi_{\text{full},i}/6}, \quad \mathbf{a}_{i,\text{HQIV}} = \frac{|\mathbf{a}_{i,N}|}{f_i} \hat{\mathbf{a}}_{i,N}.$$

The readout $\varphi_{\text{full},i} = \varphi_{\text{hom}}/(1 + |\mathbf{r}_i|/R_{\text{lapse},i}) + 6|\mathbf{a}_{i,N}|\epsilon_i$ uses the stellar co-spin Doppler lapse $\epsilon_i = 2(v_{\text{spin},i}/c) |\hat{\mathbf{v}}_i \cdot \hat{\mathbf{t}}_i|/D_R$ with $v_{\text{spin},i} = |\omega_i|R_i$. No fitted external acceleration scale enters. The scripts `hqiv_wide_binary.py` and `hqiv_wide_binary_catalog.py` integrate the coupled system with RK4, load the 36-system Chae (2026) CSVs, and expose a `--full-treatment` mode. The reference system **chae2026_58** (Gaia 5607190344506642432 / 5607189485513198208; HARPS RV; Chae $v_{\text{obs}}/v_{\text{esc},N} = 0.35$) uses projected separation $r_{\text{obs}} = 1.23 \times 10^4$ AU at $d = 38.1$ pc, $g_N = 7.9 \times 10^{-11} \text{ m s}^{-2}$, and Chae-scaled tangential speed $v = 0.19 \text{ km s}^{-1}$. The instantaneous HQIV $\gamma_{\text{eff}} \equiv a_{\text{HQIV}}/a_N$ is ≈ 1.000006 on the primary (ppm-level screen, not $\gamma \approx 1.6$). Vis-viva gives $a \approx 7.0 \times 10^3$ AU, $P \approx 4.1 \times 10^5$ yr for the same inputs.

Preset	a [AU]	e	P [yr]	$(a_{\text{HQIV}}/a_N - 1)_{\text{apo}}$
demo_circular_1au	1	0.0	0.71	3.2×10^{-6}
demo_wide_eccentric	5000	0.4	2.4×10^5	8.1×10^{-6}
literature_scale_10kau	10000	0.3	6.9×10^5	6.0×10^{-6}

The apastron column quotes the fractional boost at the slower, lower-acceleration leg; the periastron boost is slightly smaller at the same eccentricity. At 10^4 AU the Newtonian binding acceleration is $\sim 2.5 \times 10^{-10} \text{ m s}^{-2}$, still well above the homogeneous scale $\varphi_{\text{hom}}/6 \sim 2.3 \times 10^{-10} \text{ m s}^{-2}$, so the correction remains percent-level in $1 - f$ rather than a MOND-like factor-of-two shift. Running

--run-all-chaе reproduces the envelope statistics in §11.4. A decisive comparison with [?, ?] still requires matching their 3D orbital posteriors (semi-major axis, eccentricity, inclination) rather than instantaneous projected geometry alone; the catalog loader is structured for that extension.

15. OPEN ITEMS

- **Lean orbit theorems.** Bundle the chart-level orbit hypotheses (`OMaxwellFluidChartHypothesis` style) so the derived λ , the horizon repartition, and the G_{eff} time-factor application are theorems rather than calculator conventions.
- **Wide-binary posteriors.** Import per-system MCMC orbital elements from [?, ?] into `hqiv_wide_binary_catalog.py` and compare HQIV $\gamma_{\text{eff}}(t)$ along forward-modelled 3D trajectories (instantaneous screen vs. Chae γ are different observables until this is done).
- **Galaxy horizon channel.** Port flyby horizon repartition (18)–(19) to `hqiv_sparc_rotation.py` or document explicitly why disks omit it.
- **WHIM shell indices.** Tie `m_ism/m_whim` boundary amplitudes to propagation-shell coarseness from [?], not ad-hoc lock-in steps.
- **Asymptotic shell.** Replace the fixed $r_{\text{escape}} = 40 R_{\oplus}$ truncation by an adaptive condition $r\dot{v}_{\infty}/c < \epsilon$ to push absolute classical $|\Delta v|$ below mm/s without sacrificing wall-clock.
- **Per-mission geometry reconstruction.** The calculator now has a SPICE/JPL Sun–Moon tide backend. The remaining mission-grade step is replacing the approximate impact-plane azimuth fits with SPICE/JPL-derived $(\delta_{\text{in}}, \delta_{\text{out}}, \text{az})$ for each spacecraft trajectory. This is the cleanest way to either resolve or sharpen the Rosetta-I sign test (§11.4): a sharp, signed HQIV prediction that becomes decisive once geometry uncertainty is reduced, not a fitted anomaly.
- **Mission-grade density model.** The calculator implements the exponential drag posterior used in §11. A future MSIS density backend should replace the current prior envelope before claiming mission-grade atmospheric closure for NEAR or Galileo I.
- **SPARC figure refresh.** Keep `artifacts/sparc_hqiv_catalog.json` and Figure 1 in sync when SPARC data, WHIM blend, or filament cross-match conventions change.
- **Local-volume filaments.** Extend the $|\text{Dec}| \geq 55^\circ$ fallback beyond projected Carrón spines (e.g. Cosmicflows-3 / 2M++ velocity-field priors) for IC2574-class outliers still on proxy spines.

APPENDIX A. LEAN MODULE INDEX

Orbital flyby scaffold:

[Hqiv/Physics/OrbitalFlybyScaffold.lean](#) — inertia screen, screened G_{eff} , anchor-shell lemmas.

Fluid closure scaffold:

[Hqiv/Physics/HQIVFluidClosureScaffold.lean](#).

Coronal longitudinal stress:

[Hqiv/Physics/CoronalLongitudinalStress.lean](#) — boundary φ jump and even/odd-in- I discriminants (companion papers).

Shell ladder $\varphi(m)$:

[Hqiv/Geometry/OctonionicLightCone.lean](#) — `phi_of_shell`, `alpha_eq_3_5`.

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